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AUTOMATED DIABETIC RETINOPATHY GRADING USING RESNET

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ABSTRACT

This paper presents a deep learning framework for the classification of diabetic retinopathy (DR) grades from fundus images. The proposed framework is composed of three stages. First, the fundus image is preprocessed using intensity normalization and augmentation. Second, the pre-processed image is input to a ResNet Convolutional Neural Network (CNN) model in order to extract a compact feature vector for grading. Finally, a classification step is used to detect DR and determine its grade (e.g., mild, moderate, severe, or Proliferative Diabetic Retinopathy (PDR)). The proposed framework is trained using the challenging ISBI'2018 Indian Diabetic Retinopathy Image Dataset (IDRiD). To remove the training bias, the data is balanced to ensure that each DR grade is represented with the same number of images during the training process. The proposed system shows an improved performance with respect to the related techniques using the same data, evidenced by the highest overall classification accuracy of 86.67%.

Keywords: retinal, diabetic retinopathy, grading, deep learning

I. INTRODUCTION

According to the World Health Organization (WHO), diabetic retinopathy DR accounted for an estimated number of 5 million blind cases, representing around 5% of world blindness [1]. Early detection can prevent the occurrence of blindness. This paper presents a deep learning framework for the early detection and grading of DR.

DR grades are characterized by a number of retinal lesions, such as microaneurysms, hemorrhages, and hard exudates [2]. In the literature, different methods, including deep learning techniques, have been developed for lesions detection and DR grading, showing good performance. For example. Khojasteh et al. [3] built a CNN model, composed of 12 layers for detecting three lesions related to DR, i.e., exudates, hemorrhages, and microaneurysms. Lam et al. [4] applied an algorithm for localizing lesions using different CNN architectures, i.e., AlexNet, VGG16, GoogLeNet, ResNet, and Inception-v3. Eftekhari et al. [5] built a CNN model that has ten layers for the detection of microaneurysms lesions. Garcia et al. [6] developed a method for DR detection based on data preprocessing followed by comparing the performance of different CNN models (AlexNet, VGGNet16, etc.). Yang et al. [7] used a CNN model to classify DR stages to either normal, mild, moderate, or severe grade. Pratt et al. [8] used a CNN architecture for DR grading using pre-processing and data augmentation. Qummar et al. [9] used different pre-processing steps and an ensemble of CNN models, i.e., Resnet50, Inceptionv3, Xception, Dense121, and Dense169, to classify DR into relevant grades. Kori et al. [10] applied preprocessing followed by an ensemble, composed of five different Resnets and three different Densents, for DR grading. Their system had achieved an accuracy of 83.9%. Although the developed deep learning techniques have shown success in DR grading, further investigations are needed to improve the grading accuracy.

This work proposes a framework for DR grading based on the deep learning ResNet-50 model and smart data preprocessing, using the principles of data augmentation and data balancing, to improve the quality of the training process. The proposed framework shows an improved accuracy for DR grading over other related state-of-the-art methods on the same data. This paper is organized as follows. Section II illustrates the proposed framework. Section III discusses the framework results. Finally, section IV includes the paper conclusions and future works.

II. METHODS

As shown in Figure 1, the proposed framework consists of three pipelined stages: a preprocessing stage that applies intensity normalization, vertical flipping augmentation, and data balancing, a feature extraction stage that utilizes a ResNet CNN model, and a classification stage that uses either a Supported Vector Machine (SVM) classifier or a Neural Network (NN) pixel-wise classier.

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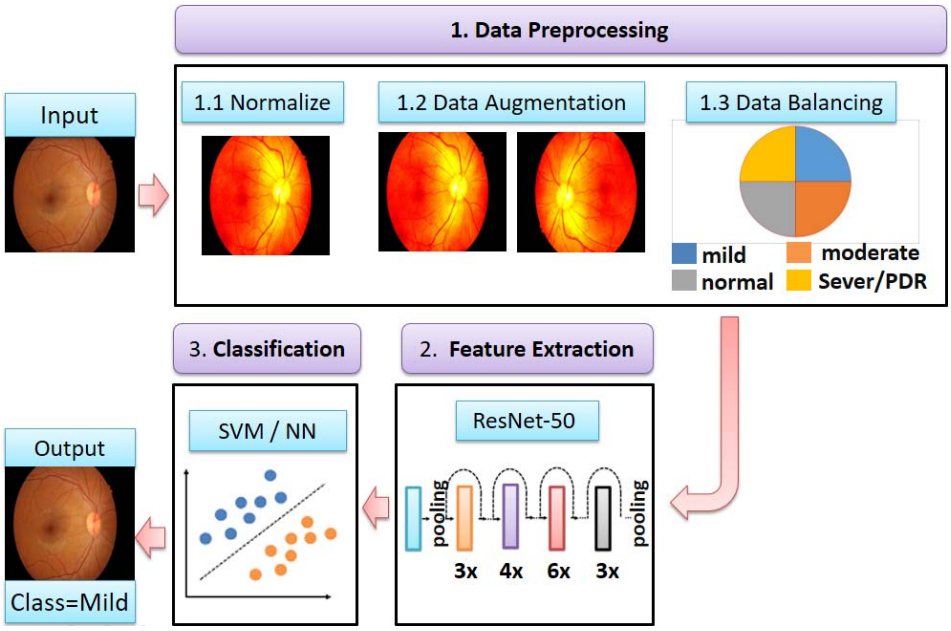


Fig. 1: The proposed framework consists of a preprocessing stage, a feature extraction stage and classification stage.

A. Collected Database (IDRiD)

The data is collected from the ISBI'2018 challenging Indian Diabetic Retinopathy Image Dataset (IDRiD). The IDRiD is composed of 516 fundus retinal images, with five class labels (i.e., normal, mild, moderate, sever, and Proliferative Diabetic Retinopathy (PDR)). Figure 2 shows two typical examples from the IDRiD data for each class. Data are captured using an alpha Kowa VX-10 digital camera, with a Field of View (FOV) of 50 degrees. Specifications of original images are as follows: 2848×4288 pixel resolution and "jpg" file format. More details about the database can be found in [11].

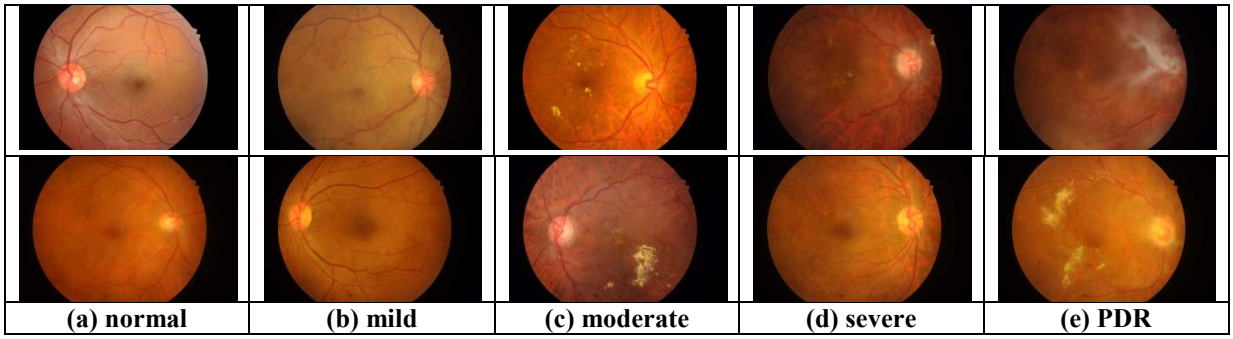


Fig. 2: Examples of typical fundus images from the IDRiD database that represent different DR stages: (a) normal, (b) mild, (c) moderate, (d) sever, and (e) PDR

B. Image Preprocessing

Image prespocessing is implemented through three steps; intensity normalization, augmentation, and data balancing (see Fig. 1). First, the image intensities are normalized to be between 0 and 1, as follows:

$$Im(x,y)_{norm} = \frac{Im(x,y) - min(Im)}{max(Im) - min(Im)} \quad (1)$$

Where Im denotes the input image, $Im(x,y)$ denotes the intensity of the pixel located at the Cartesian coordinates (x,y), $min(.)$ performs the minimum function, $max(.)$ performs the maximum function, and $Im(x,y)_{norm}$ denotes the output normalized pixel intensity level between 0 and 1. A further normalization step is applied such that all normalized images have zero means (μ) and unit standard deviations (σ), using Eq. 2:

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$$Im_{z-norm} = \frac{Im_{norm} - \mu}{\sigma} \quad (2)$$

Where Im_{z-norm} denotes the final normalized image with zero mean ($\mu = 0$) and unit variance ($\sigma = 1$).

After all images are normalized, they have been augmented, in order to increase the training data, and therefore the quality of the training process. The augmentation is done by flipping each input image along the vertical axis. The visualization of the output images of the proposed prepocessing steps (data normalization, augmentation) are shown in Figure 3.

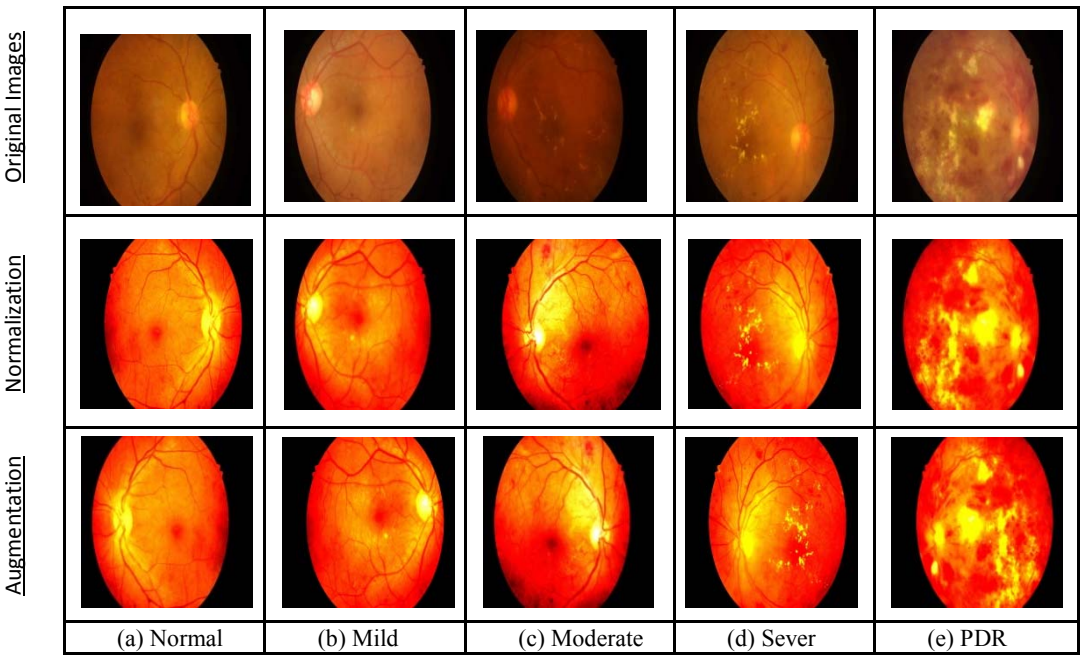


Fig. 3: Data pre-processing and augmentation processes. The first row shows five original images, each from a different DR grade (i.e., (a) normal, (b) mild, (c) moderate, (d) sever, and (e) PDR). The second row shows their normalization images with zero mean and unit variance. The third row shows their vertically-flipped augmented images

In order to efficiently train the utilized deep learning CNN model, a data balancing step is performed, in order to provide the CNN model, during the training phase, with an equally split of data per each DR grade. This can assist in removing any bias during the training process. Therefore, the proposed framework trains the CNN model with an equal number of images, selected randomly, for each DR grade.

C. Deep Learning CNN Model

The proposed framework applies a transfer learning using the pertained Res-Net50 [12] CNN model, consisting of 50 weighted layers. Res-Net50 has four stages, each stage consists of three convolutional layers, and is replicated n times (see Figure 4). Res-Net stands for a residual network, where blocks of convolutional layers are skipped by using shortcut connections. This feature can assist the ResNet model to learn the data specific global features. The detailed structure of the network is shown in Figure 4. To apply the transfer learning [13], the parameters of the convolutional layers are transferred without changes, and the fully connected layer (FC1000 layer) is replaced with a shallow classifier with four class labels, representing the DR grades. The shallow classifier is trained using the target data (IDRiD) labels.

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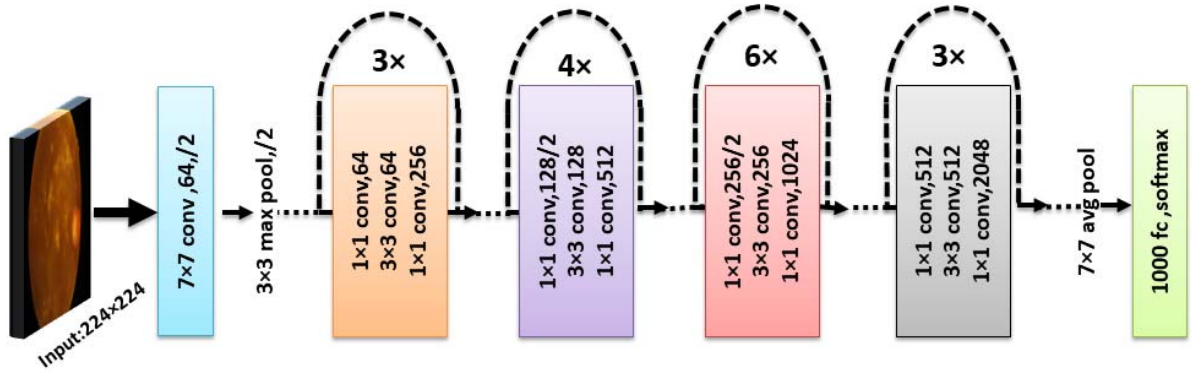


Fig. 4: ResNet-50 architecture developed by He et al. [12]. ResNet-50 has 4 stages, each stage consists of three convolutional layers and is replicated *n* times (denoted as *n*× in the figure)

D. First-Stage Classification

In this step, we investigated two types of shallow classifiers to classify a DR image. The first classifier is based on a pixel-wise feedforward Neural Network (NN) classifier, composed of one layer of fully connected weights between the feature vector of the Resnet50 model and the output layer composed of a number of nodes equals to the DR grades. The second one performs a binary linear kernel Supported Vector Machine (SVM) [14] classification, based on the selected features. To train the first-stage system classifiers, four class labels are used: 'normal', 'mild', 'moderate', and 'severe/PDR'. Where, the DR grades 'Severe' and 'PDR' have been grouped into one label "Severe/PDR" during all experimentation, due to the closeness of their images to each other. To select the best classifier, the proposed system has been tested using a standard performance evaluation metric, i.e., the classification accuracy [15].

E. Second-Stage classification

Due to the closeness of the two classes (Severe and PDR), the original Resnet is not able to distinguish between them among the DR grades. To afford better accuracy, a second stage Resent model is added. This stage is learned offline to distinguish only between the two classes (Severe and PDR), where only corrected Severe/PRD images of eh first stage are input to it. The output of this stage is either "Severe" or "PDR".

F. Performance Evaluation

To evaluate the performance of the proposed system, a standard performance evaluation metric, i.e., the classification accuracy (CA) [15] is used. CA_g represents the fraction of the correctly classified images for the specific grade "g", defined as follows [16]:

$$CA_g = \frac{\text{Number of correctly classified grade } g \text{ images}}{\text{Total number of gade } g \text{ images}} \qquad (3)$$

where g refers to the grade of the DR, i.e., $g \in$ {"normal", "mild", "moderate", "severe/PDR"}. For example, if 90 out of a total of 100 normal images are correctly classified, then CA_{normal}=90%. The overall system classification accuracy CA can be found as the average of the individual classification accuracies for each grade.

III. Results and Discussion

The detailed setup of the proposed system, related results and discussions are illustrated below.

A. Experimental Setup

To train the Resnet CNN models, the data is split randomly into 70% training dataset and 30% testing dataset. The training process minimizes the cross-entropy loss, using a learning rate of 10^{-4} and a maximum number of epochs of 20.

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B. First-Resenet Stage Results

To investigate the potential of each preprocessing step (i.e., contrast normalization, data augmentation, and data balancing) in the proposed framework performance, results are compared, for the first stage classification, between six different cases:

- **Case A:** no preprocessing steps (using the raw data)
- **Case B:** only normalization is applied (one step)
- **Case C:** normalization and augmentation (two-steps)
- **Case D:** normalization and balancing (two-steps)
- **Case E:** applying only Eq. 1 normalization, augmentation, and balancing (three-steps)
- **Case F:** applying full normalization (both Eq. 1 and Eq. 2), augmentation, and balancing (three-steps)

For each case, results are compared between using a NN pixel-wise classifier or a SVM classifier. As shown in Table 1, the NN pixel-wise classifier provides a better accuracy than a SVM classifier for all the investigated cases (i.e., case A, B, C, D, E, and F). As shown in Table 1 and illustrated in Fig. 5, the system performance gradually increases by increasing the number of the utilized preprocessing steps. For example, using the proposed intensity normalization step improves the overall CA from 62.34% to 62.99%. Adding an augmentation or a balancing step increases the performance to CA=68.96% or CA=69.44%, respectively. Finally, adding all the proposed steps improve the proposed system accuracy to CA=84.72% or CA=88.00%, if the first step applies only Eq. 1 or applies both Eq.1 and Eq. 2, respectively. These results shows the impact of the proposed preprocessing steps for improving the performance.

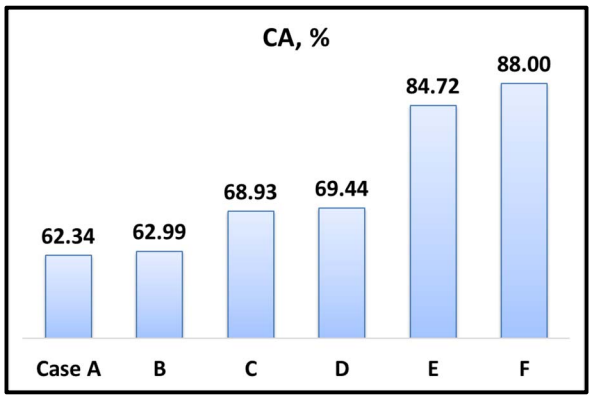


Fig. 5: Results of the first-stage Resent comparing the different six cases

C. Second-Resent Stage Results

A second stage Resent model is added to distinguish between the two classes (Severe or PDR), where only corrected Severe/PRD images of the first stage are input to it. To highlight the advantage of the proposed two-stage system for DR grading, comparison with other related work [10] on the same challenging IDRiD data. As shown in Table 2, the proposed two-stage Resent system has shown a superior performance through applying all the proposed preprocessing steps.

D. Complexity and Time Performance Analysis

The advantage of the proposed two-Resnet system is its ability to distinguish between all the classes with high performance, evidenced by the CA value. In addition, since the training is performed offline, the overhead of adding the second stage Resnet is minimized, taking into account that the second Resnet stage is performed only if the first stage outputs a class of "Severe/PDR". Comparing with the related work, [10] applies preprocessing followed by an ensemble of eight CNN models. These models include five different sizes Resnets (Resnet18, 34, 50, 101, and 152) and three different sizes Densents (Dense121, 169, and 201). Contradictory, the proposed system applies only Resnet50 models, composed of only 50 layers. Compared to the 150 layers of largest Resent in the ensemble (Resent152) and the 201 layers of the largest Densenet in the ensemble (Dense 201), it is clear that the training of the proposed Resent-50 is of much less computational cost. Moreover, the test time will be significantly smaller. Using a 2.5 intel CORE i5 2947MCPU and 4GB of RAM and Matlab 2018a, the proposed two-Resnet system

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process a test image and provide its grade typically on an average time of 1.30 second (0.22 sec for normalization, 0.41 sec to output the first-stage DR grade, and 0.40 to output the second-stage DR grade).

Table 1: Comparison Results for first stage grading of four DR classes ('normal', 'mild', 'moderate', and 'severe/PDR') using different settings: Case A (no preprocessing), Case B (normalization), Case C (normalization & augmentation), Case D (normalization & balancing), Case E (only Eq. 1 preprocessing, augmentation & balancing), Case F (normalization, augmentation & balancing) using ResNet50 comparing neural network (NN) or (SVM) classifiers

Case	Preprocessing			Label	# Images	CA, %	
	Normal.	Aug.	Balancing			*ResNet50 +NN*	*ResNet50 +SVM*
A	NO	NO	NO	**Mild**	25	0.00	0.00
				Moderate	169	58.82	72.03
				Normal	167	80.00	55.56
				Severe/PDR	155	56.52	52.78
				Total	516	**62.34**	45.09
B	YES	NO	NO	**Mild**	25	0.00	0.00
				Moderate	169	29.41	30.51
				Normal	167	82.00	71.79
				Severe/PDR	155	89.13	82.41
				Total	516	**62.99**	46.18
C	YES	YES	NO	**Mild**	50	0.00	0.00
				Moderate	334	37.62	14.77
				Normal	338	87.00	96.58
				Severe/PDR	310	94.62	82.49
				Total	1032	**68.93**	48.46
D	YES	NO	YES	**Mild**	25	42.86	70.59
				Moderate	**25**	42.86	11.76
				Normal	**25**	71.43	5.88
				Severe/PDR	**50**	93.33	100
				Total	**125**	**69.44**	47.06
E	Only Eq. 1	YES	YES	**Mild**	50	100	93.33
				Moderate	50	66.67	20.00
				Normal	50	66.67	53.33
				Severe/PDR	100	96.30	93.33
				Total	250	**84.72**	65.00
F	YES	YES	YES	**Mild**	50	93.33	93.33
				Moderate	50	66.67	20.00
				Normal	50	100	53.33
				Severe/PDR	100	90.00	93.33
				Total	1250	**88.00**	65.00

Table 2: Comparison Results between related method ([10], 2018) for five class DR grading and the proposed two-stage system using ResNet50 comparing NN or SVM classifiers

Method	Label	# Images	CA, %	
			ResNet50 +NN	*ResNet50 +SVM*
Two-stage Proposed system (Case E)	**Mild**	50	93.33	93.33
	Moderate	50	66.67	20.00
	Normal	50	100	53.33
	Severe	50	86.67	93.33

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	PDR	50	86.67	87.33
	Total	1250	86.67	65.00
DenseNet/ResNet ensemble [10]	Total Images	56	83.90	

IV. CONCLUSION

This paper presents a three-stage framework for automated DR grading. The proposed framework is composed of preprocessing, feature extraction, and classification. Experimental results show that using data augmentation and data balancing can significantly improve the system performance. Comparison results with other related work on the IDRiD challenging data confirm the accuracy of the proposed framework for DR grading. In the future, we plan to investigate using other deep learning models to improve the performance and other databases to check the robustness of the proposed system.

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